

1 (a) Photosynthesis is a complex process involving the transfer of light energy into chemical energy.

(i) Describe the role of photosynthetic pigments.

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(ii) Name the precise location in a chloroplast of photosynthetic pigments.

..... [1]

(iii) Name a practical technique to separate photosynthetic pigments.

..... [1]

(b) The rate of photosynthesis is affected by many environmental factors.

A student carried out an experiment to investigate the relationship between the concentration of carbon dioxide available to an aquatic plant, *Cabomba caroliniana*, and its rate of photosynthesis.

Fig. 1.1 shows the experimental set-up for this investigation.

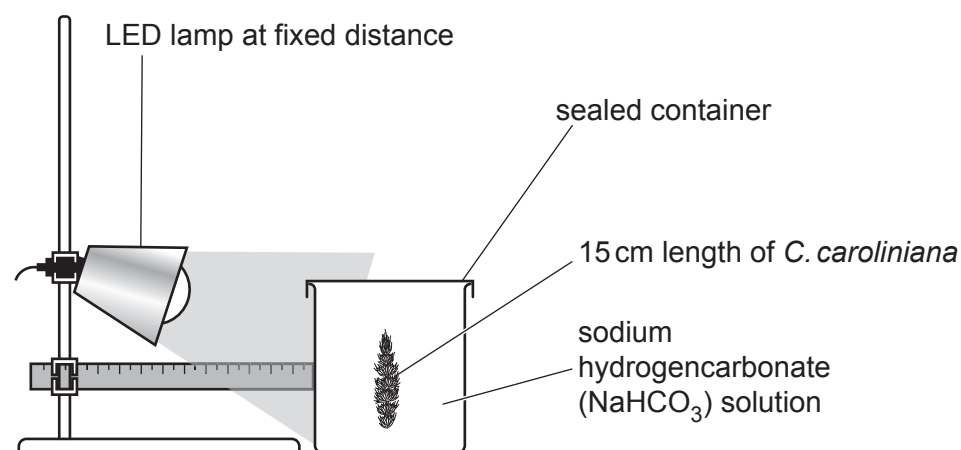


Fig. 1.1

- (ii) The percentage change in dissolved oxygen for *C. caroliniana* at 0.00 mol dm^{-3} of sodium hydrogencarbonate solution is negative.

Suggest reasons for this negative value.

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- (iii) To minimise temperature changes, the student decided to use an LED lamp as a light source. LED lamps release very little heat energy.

Explain the importance of minimising temperature changes in this experiment.

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[Total: 12]